CS 57800 – Statistical Machine Learning

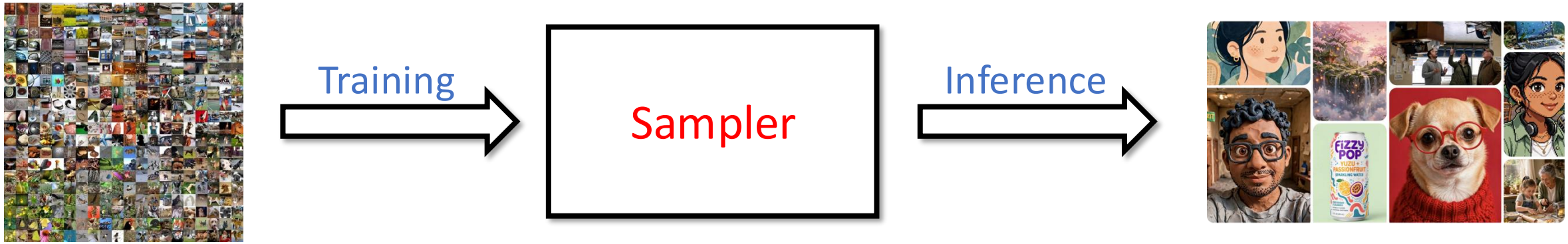
Lecture 4 Diffusion Model Theory

https://ruizhezhang.com/course_fall_2026.html

This slides is based on Sitan Chen's lecture notes.

Generative Modeling a.k.a. “Learning to Sample”

- **Input:** samples from data distribution (e.g., natural images)
- **Output:** a sampler that can generate samples from distributions that are close to the data distribution



Modern paradigm: progressive refinement

- Training: $x_0 \sim q$, “corrupt” x_0 to get x_t, corrupt more to get x_{t+h}, and train a neural network to predict x_t from x_{t+h}

Generative Modeling a.k.a. “Learning to Sample”

- **Input:** samples from data distribution (e.g., natural images)
- **Output:** a sampler that can generate samples from distributions that are close to the data distribution



Training

Sampler

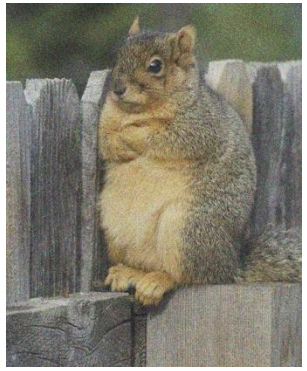
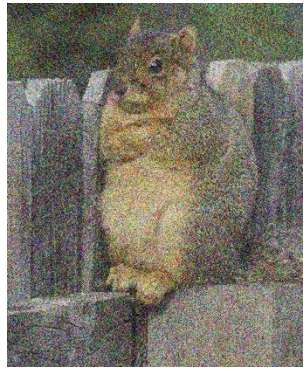
Inference

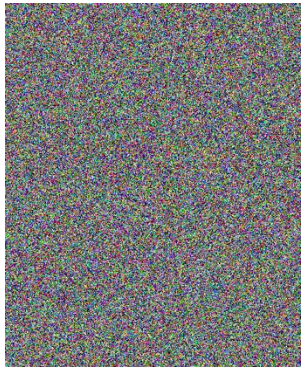


Modern paradigm: progressive refinement

- Inference: start from a “zero-information” point, progressively refine the point until it looks like a sample from the data distribution

Gaussian noising



- $x_0 \sim q$ (a sample from the data distribution)
- $x_t \sim \mathcal{N}(x_0, tI)$ a Gaussian noising
- As $t \to \infty$, x_t "forgets" the initial point x_0:

$$\mathrm{KL}(\mathcal{N}(x_0, tI) \| \mathcal{N}(0, tI)) = \frac{\|x_0\|^2}{2t}$$

- We can take a large T, sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$ as x_T, and try to "recover" x_0 (?)

Gaussian denoising

- $x_0 \sim q$ (a sample from the data distribution)
- $x_t \sim \mathcal{N}(x_0, tI)$ a Gaussian noising
- Given x_t, what is the posterior over x_0?

$$\Pr[x_0|x_t] \propto \Pr[x_t|x_0] \cdot \Pr[x_0]$$

$$\propto \exp\left(-\frac{\|x_t - x_0\|^2}{2t}\right) \cdot q(x_0)$$

- When t is tiny, this posterior distribution is nearly a Gaussian concentrated around x_t
- Hard to characterize for a general data distribution q

Gaussian denoising

- $x_0 \sim q$ (a sample from the data distribution)
- $x_t \sim \mathcal{N}(x_0, tI)$ a Gaussian noising
- Given x_t, what is the **mean** of the posterior over x_0?
- **Tweedie's formula:** let q_t be the probability density of x_t, then

$$m_t(x_t) \coloneqq \mathbb{E}[x_0|x_t] = x_t + t\nabla \log q_t(x_t)$$

Bayesian correction

- Attempt for progressive refinement: $\mathbb{E}[x_{t-h}|x_t] = x_t + h\nabla \log q_t(x_t)$
  - Sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$,
  - Compute $\tilde{x}_{T-h} \leftarrow \tilde{x}_T + h\nabla \log q_T(\tilde{x}_T)$
  - Compute $\tilde{x}_{T-2h} \leftarrow \tilde{x}_{T-h} + h\nabla \log q_{T-h}(\tilde{x}_{T-h})$
  - …
  - Output $\tilde{x}_0$

Gaussian denoising

- Recall that

$$\Pr[x_{t-h}|x_t] \propto \exp\left(-\frac{\|x_t - x_{t-h}\|^2}{2h}\right) \cdot q_{t-h}(x_{t-h})$$

- Most of the probability mass is around $O(\sqrt{h})$ away from x_t
- Tweedie's formula gives a correction $h\nabla \log q_t(x_t)$ of order h
- We can show that $\text{law}(x_{x-h}|x_t) \approx \mathcal{N}(x_t + h\nabla \log q_t(x_t), tI)$
  - Sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$,
  - Sample $\tilde{x}_{T-h} \sim \mathcal{N}(\tilde{x}_T + h\nabla \log q_T(\tilde{x}_T), hI)$
  - Sample $\tilde{x}_{T-2h} \sim \mathcal{N}(\tilde{x}_{T-h} + h\nabla \log q_{T-h}(\tilde{x}_{T-h}), hI)$
  - …
  - Output $\tilde{x}_0$

Diffusion sampling

- Sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$,
- Sample $\tilde{x}_{T-h} \sim \mathcal{N}(\tilde{x}_T + h\nabla \log q_T(\tilde{x}_T), hI)$
- Sample $\tilde{x}_{T-2h} \sim \mathcal{N}(\tilde{x}_{T-h} + h\nabla \log q_{T-h}(\tilde{x}_{T-h}), hI)$
- …
- Output $\tilde{x}_0$

How to get $\nabla \log q_t$, the **score** of q_t?

$$m_t(x_t) \coloneqq \mathbb{E}[x_0|x_t] = x_t + t\nabla \log q_t(x_t)$$

- Score estimation: train a neural network on (x_0, t, x_t):

$$m_t(\cdot) \coloneqq \arg\min_f \|x_0 - f(x_t)\|^2$$

Diffusion sampling

An alternative approach

- Sample $\tilde{x}_T \sim \mathcal{N}(0, TI)$,
- Compute $\tilde{x}_{T-h} = \tilde{x}_T + \frac{1}{2} h \nabla \log q_T(\tilde{x}_T)$
- Compute $\tilde{x}_{T-2h} = \tilde{x}_{T-h} + \frac{1}{2} h \nabla \log q_{T-h}(\tilde{x}_{T-h})$
- …
- Output $\tilde{x}_0$

Diffusion sampling

We made several assumptions/heuristics:

- $\tilde{x}_T \sim \mathcal{N}(0, TI)$ is pretended to be x_T
- $\text{law}(x_{x-h}|x_t) \approx \mathcal{N}(x_t + h\nabla \log q_t(x_t), tI)$
- h is not set to be infinitesimally small
- Score estimation with finite data / compute

Stochastic calculus is the mathematical language to study generative modeling in a continuous space and continuous time

Brownian motion

$(\boldsymbol{B}_t)_{t\geq 0} \subset \mathbb{R}^d$ is a stochastic process (random sequence of vectors indexed by t) such that:

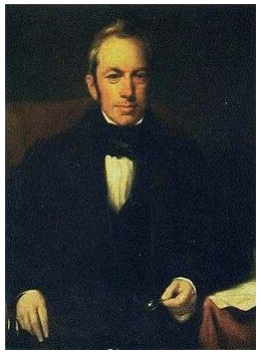
- Starts at origin: $\boldsymbol{B}_0 = 0$ is the origin in $\mathbb{R}^d$
- Continuous paths: With probability 1, $t \mapsto \boldsymbol{B}_t$ is continuous
- Independent increments: For all $0 = t_0 < t_1 < \cdots < t_k$, the random vectors $\boldsymbol{B}_{t_{i+1}} - \boldsymbol{B}_{t_i}$ for $0 \leq i < k$ are mutually independent
- Gaussian increments: For all $0 \leq s \leq t$, $\boldsymbol{B}_t - \boldsymbol{B}_s \sim N(0, (t-s)I)$

The probability space that Brownian motion is defined on is denoted $\{\mathcal{F}_t\}_{t\geq 0}$, a filtration satisfying $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $0 \leq s \leq t$. We say that Brownian motion is a stochastic process adapted to the filtration $\{\mathcal{F}_t\}_{t\geq 0}$

$\mathcal{F}_t$ contains the information of the randomness used up to time t

Brownian motion

1827 — 1900 — 1905 — 1908 — 1923



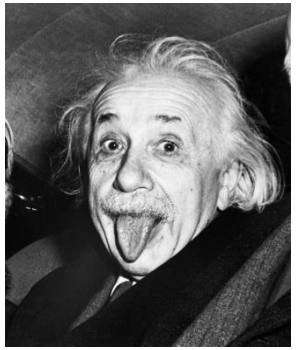
Robert Brown

Observes erratic motion of pollen grains in water under microscope



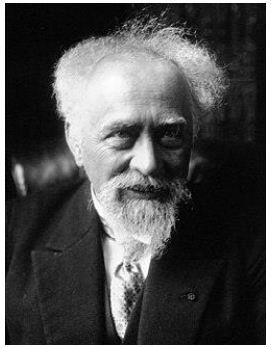
Louis Bachelier

First mathematical treatment of Brownian motion, used to model stock prices



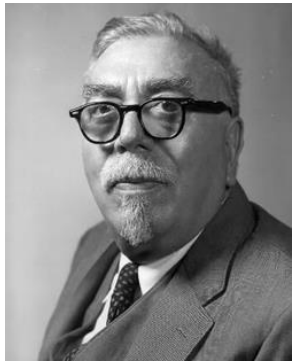
Albert Einstein

Explains *physical* mechanism behind Brownian motion, used to estimate Avogadro's number



Jean Perrin

Experimentally confirms Einstein's predictions, receives Nobel Prize

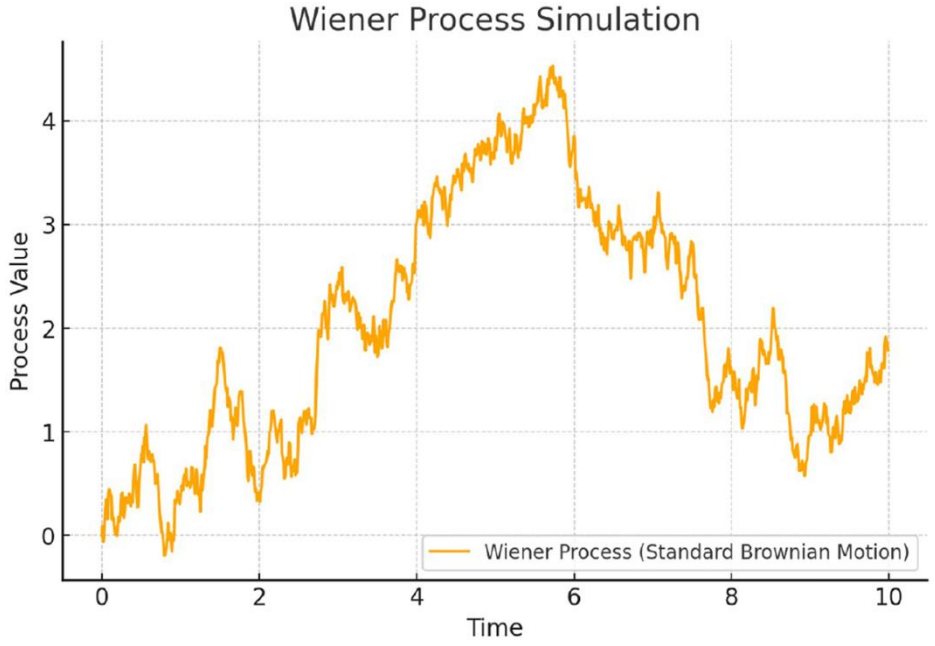


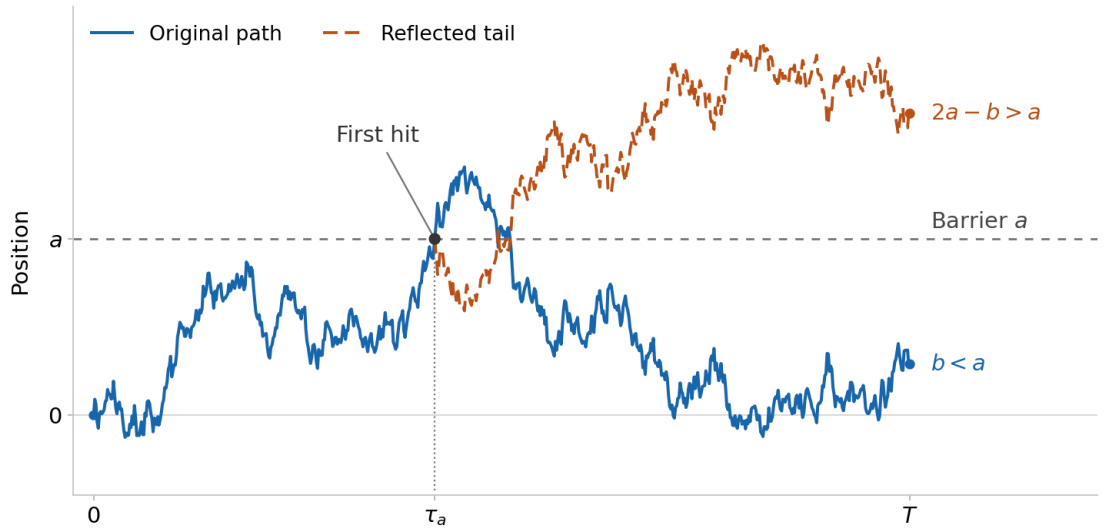
Norbert Wiener

Develops fully rigorous mathematical theory for Brownian motion

Brownian motion

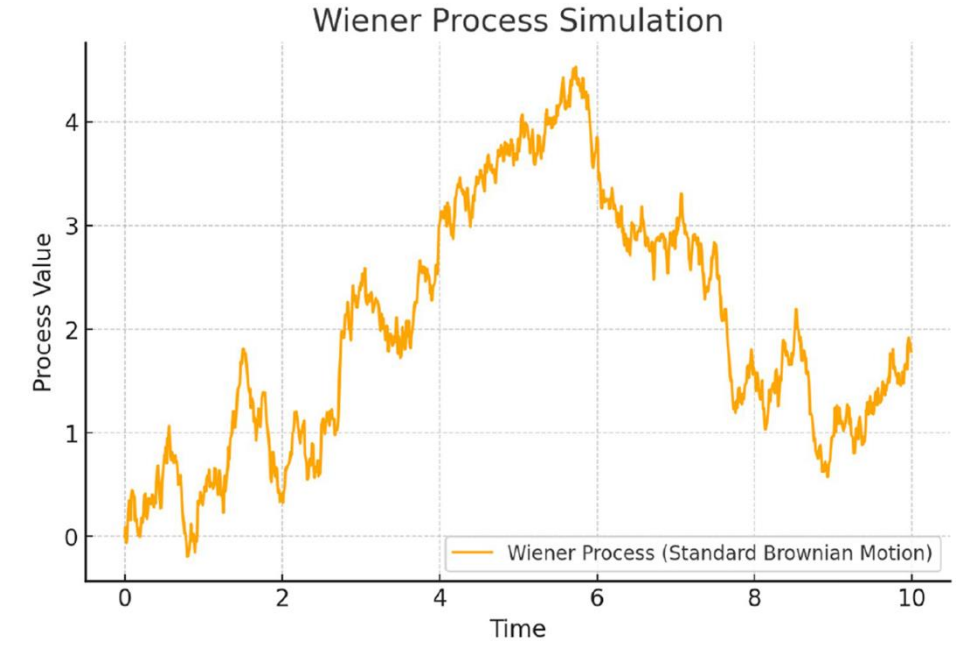
- Over a time window t, $\boldsymbol{B}_t$ moves $O\left(\sqrt{td}\right)$
- Although is $t \mapsto \boldsymbol{B}_t$ is a.s. continuous, it's a.s. nowhere differentiable
- Induces a probability measure over "path space" $[0,T]\times\mathbb{R}^d$ (Wiener measure)
- The Wiener measure is unique, but pathwise realization is **not** unique
  - Suppose $\boldsymbol{B}_t$ is some realization
  - $-\boldsymbol{B}_t$ is also a Brownian motion
  - Reflection principle: At the first time $\boldsymbol{B}_t$ hits a hyperplane H, reflect $\boldsymbol{B}_t$ about H

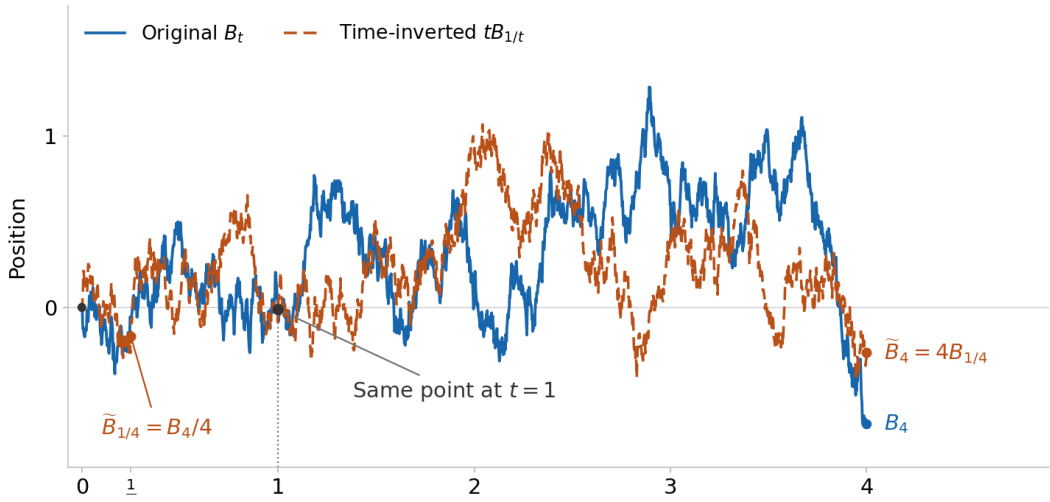




Brownian motion

- Over a time window t, $\boldsymbol{B}_t$ moves $O\left(\sqrt{td}\right)$
- Although is $t \mapsto \boldsymbol{B}_t$ is a.s. continuous, it's a.s. nowhere differentiable
- Induces a probability measure over "path space" $[0,T]\times\mathbb{R}^d$ (Wiener measure)
- The Wiener measure is unique, but pathwise realization is **not** unique
  - Suppose $\boldsymbol{B}_t$ is some realization
  - $-\boldsymbol{B}_t$ is also a Brownian motion
  - Reflection principle: At the first time $\boldsymbol{B}_t$ hits a hyperplane H, reflect $\boldsymbol{B}_t$ about H
  - Time inversion: $t\boldsymbol{B}_{1/t}$





Stochastic integrals

Given continuous process $(\mathbf{H}_t)_{t\geq 0} \in \mathbb{R}^{d\times d}$ adapted to the filtration generated by $(\boldsymbol{B}_t)_{t\geq 0}$, the Itô integral

$$\mathbf{x}_t = \int_0^t \mathbf{H}_s \, \mathrm{d}\boldsymbol{B}_s$$

is the stochastic process whose value at time t is the random vector given by taking the probability limit over meshes $P = \{t_k\} \subset [0, t]$ such that $0 = t_0 < \cdots < t_{|P|} = t$:

$$\int_0^t \mathbf{H}_s \, \mathrm{d}\mathbf{B}_s = \lim_{\|P\|_{\text{gap}}\to 0} \sum_{j=1}^{|P|} \mathbf{H}_{t_j} \cdot \left(\boldsymbol{B}_{t_{j+1}} - \boldsymbol{B}_{t_j}\right)$$

where $\|P\|_{\text{gap}} \coloneqq \max_{j\in[|P|]} \left|t_j - t_{j-1}\right|$

Stochastic integrals

The Itô integral $\mathbf{x}_t = \int_0^t \mathbf{H}_s \, \mathrm{d}\boldsymbol{B}_s$ is a continuous martingale

- Continuity: With prob. 1 over the randomness, $t \mapsto \mathbf{x}_t$ is continuous
- Martingale: $\mathbb{E}[\mathbf{x}_t | \mathcal{F}_s] = \mathbf{x}_s \iff \mathbb{E}\left[\int_s^t \mathbf{H}_s \, \mathrm{d}\boldsymbol{B}_s \,\middle|\, \mathcal{F}_s\right] = 0$
- Itô isometry:

$$\mathbb{E}\left[\left\|\int_0^t \mathbf{H}_s \, \mathrm{d}\boldsymbol{B}_s\right\|^2\right] = \mathbb{E}\left[\int_0^t \|\mathbf{H}_s\|_F^2 \, \mathrm{d}s\right]$$

$\mathbb{E}[(a_1 g_1 + a_2 g_2 + \cdots + a_k g_k)^2] = a_1^2 + \cdots + a_k^2$ for $g_1, \ldots, g_k \sim N(0,1)$

"$\mathrm{d}\boldsymbol{B}_t^2 = \mathrm{d}t$" (because $\boldsymbol{B}_{t+\mathrm{d}t} - \boldsymbol{B}_t \sim \mathcal{N}(0, \mathrm{d}t)$)

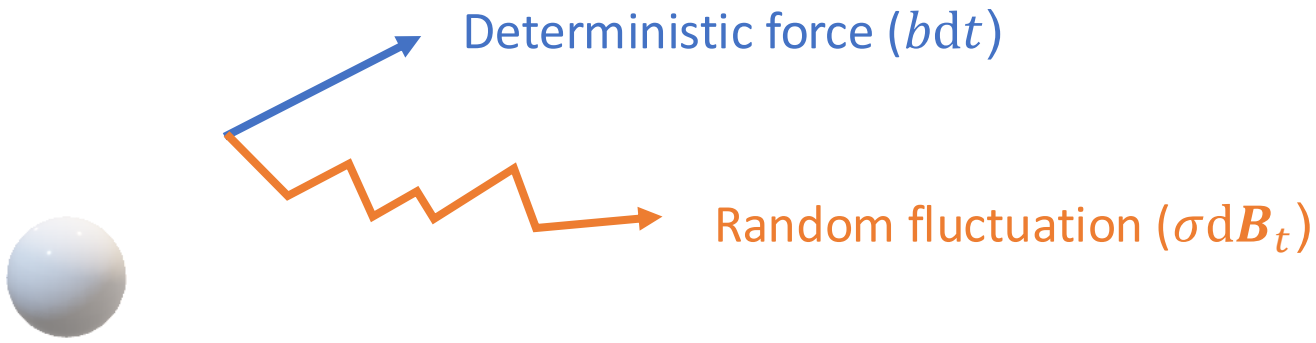
Drift-diffusion processes

A drift-diffusion process $\{\mathbf{x}_t\}_{t\geq 0}$ on $\mathbb{R}^d$ is driven by a time-dependent drift $b_t\colon \mathbb{R}^d \to \mathbb{R}^d$ and a diffusion coefficient $\sigma_t\colon \mathbb{R}^d \to \mathbb{R}^{d\times d}$, and captured by the stochastic differential equation (SDE):

$$\mathrm{d}\mathbf{x}_t = b_t(\mathbf{x}_t)\mathrm{d}t + \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$

We can write this stochastic process in integral form (Itô process):

$$\mathbf{x}_t = \mathbf{x}_0 + \int_0^t b_t(\mathbf{x}_t)\mathrm{d}t + \int_0^t \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$



Drift-diffusion processes

A drift-diffusion process $\{\mathbf{x}_t\}_{t\geq 0}$ on $\mathbb{R}^d$ is driven by a time-dependent drift $b_t\colon \mathbb{R}^d \to \mathbb{R}^d$ and a diffusion coefficient $\sigma_t\colon \mathbb{R}^d \to \mathbb{R}^{d\times d}$, and captured by the stochastic differential equation (SDE):

$$\mathrm{d}\mathbf{x}_t = b_t(\mathbf{x}_t)\mathrm{d}t + \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$

We can write this stochastic process in integral form (Itô process):

$$\mathbf{x}_t = \mathbf{x}_0 + \int_0^t b_t(\mathbf{x}_t)\mathrm{d}t + \int_0^t \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$

- Technically need to check such a process exists and is uniquely defined

Theorem. If b_t and σ_t have a finite Lipschitz constant uniformly over $t \in [0, T]$, then the SDE solution exists and is unique

Drift-diffusion processes

Ornstein-Uhlenbeck process

$$\mathrm{d}\mathbf{x}_t = -\mathbf{x}_t \mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t$$

- Without the diffusion term, the ODE $\mathrm{d}x_t = -x_t \mathrm{d}t$ has solution $x_t = e^{-t}x_0$
- We can solve this SDE in a similar way as solving the ODE:

$$\mathrm{d}(e^t \mathbf{x}_t) = e^t \mathbf{x}_t \mathrm{d}t + e^t \mathrm{d}\mathbf{x}_t = e^t \mathbf{x}_t \mathrm{d}t + e^t\left(-\mathbf{x}_t \mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t\right) = \sqrt{2}e^t d\boldsymbol{B}_t$$

$$\mathbf{x}_t = e^{-t}\mathbf{x}_0 + \underbrace{\int_0^t \sqrt{2}e^{-(t-r)}\mathrm{d}\boldsymbol{B}_r}$$

Distributed as $\mathcal{N}(0, \beta_t^2 I)$ with $\beta_t^2 = 1 - e^{-2t}$

- Solution: $\mathbf{x}_t \sim \mathcal{N}(e^{-t}\mathbf{x}_0, (1 - e^{-2t})I)$
- As $t \to \infty$, $\mathrm{Law}(\mathbf{x}_t) \to \mathcal{N}(0, I)$, i.e. the standard Gaussian is the stationary distribution

Drift-diffusion processes

(Overdamped) Langevin diffusion

For a potential / Hamiltonian $V\colon \mathbb{R}^d \to \mathbb{R}$, define

$$\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t$$

"Gradient descent with noise"

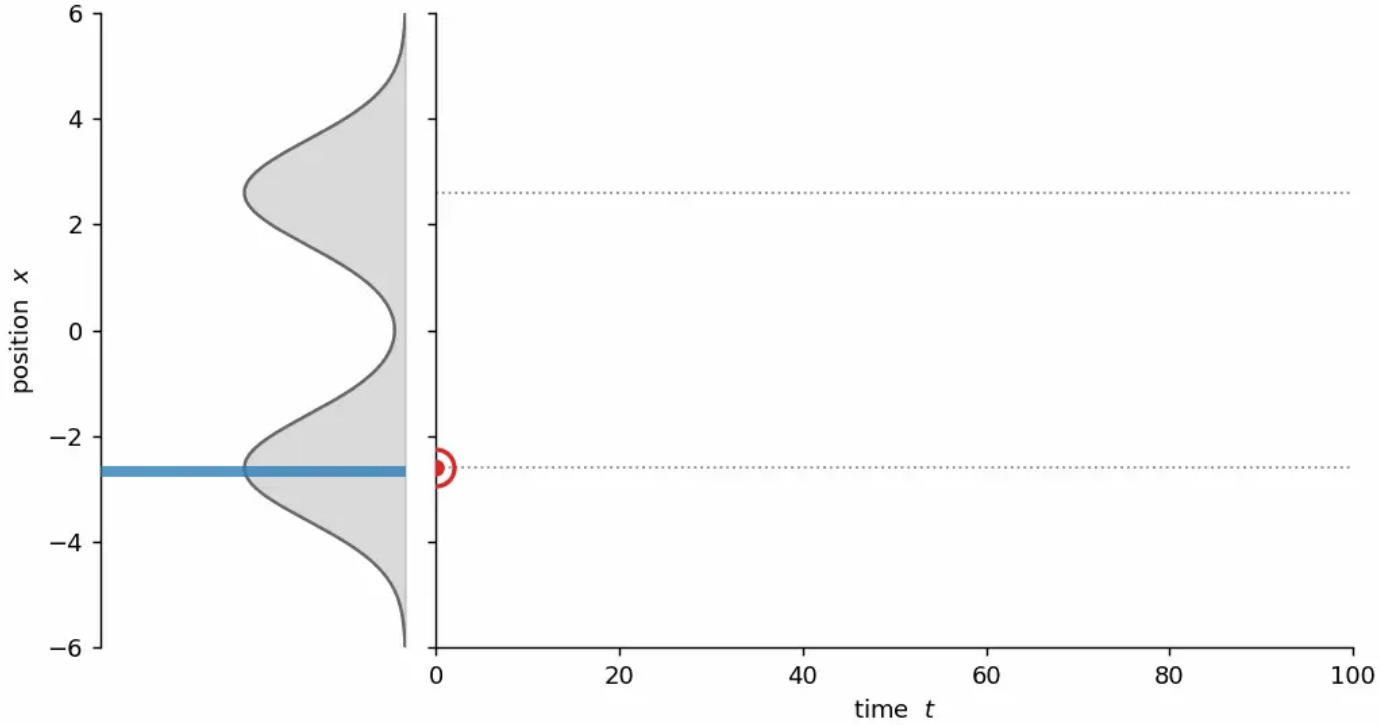
- Stationary distribution: as $t \to \infty$, $\mathrm{Law}(\mathbf{x}_t) \to \pi(x) = \frac{1}{Z}e^{-V(x)}$, i.e. the Gibbs distribution
- The drift $-\nabla V$ is the score function of the target π
- OU is a special case with $V(x) = \frac{1}{2}\|x\|^2$

Sampling from Gibbs Distributions

$$\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t$$

Euler–Maruyama discretization:

$$x_{t+h} = x_t - h\nabla V(x_t) + \sqrt{2h}\gamma, \qquad \gamma \sim \mathcal{N}(0, I)$$

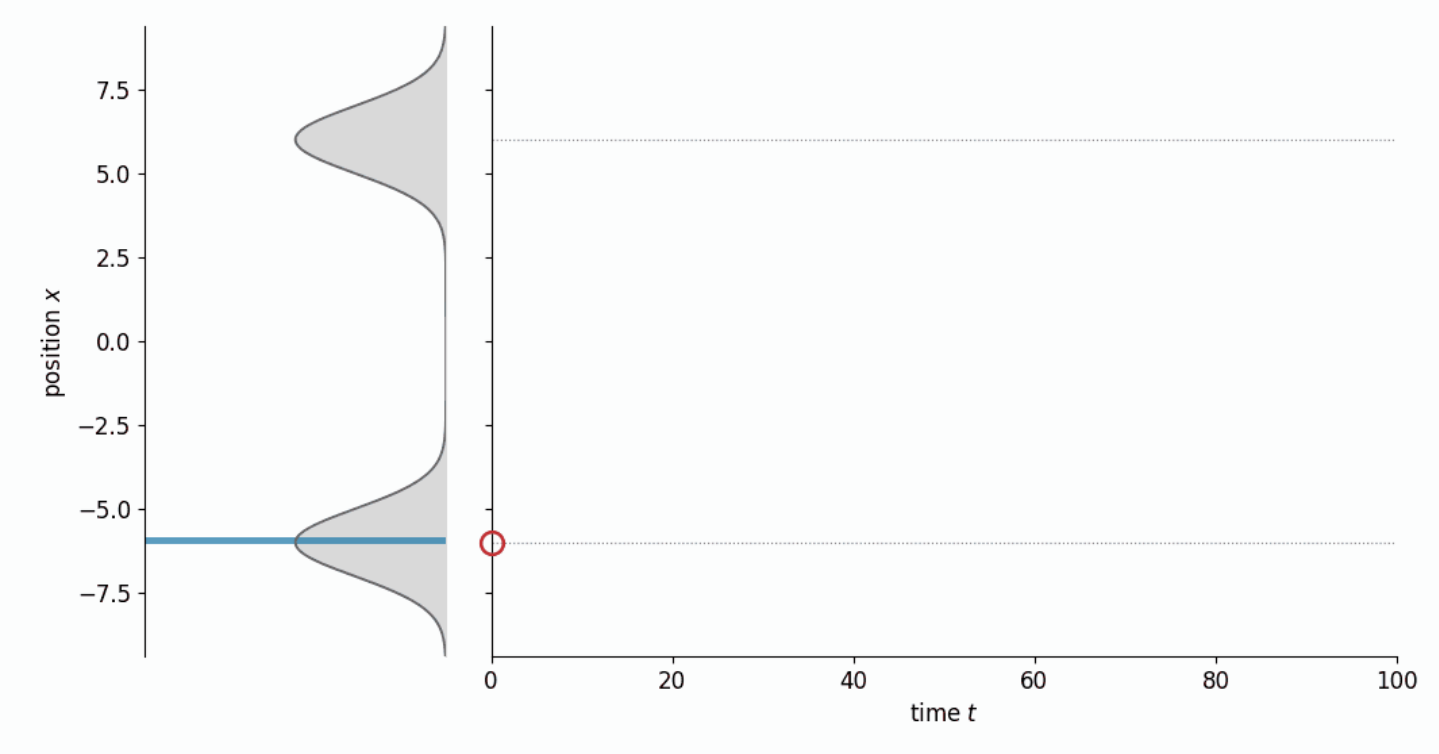


Sampling from Gibbs Distributions

$$\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\mathrm{d}t + \sqrt{2}\mathrm{d}\boldsymbol{B}_t$$

Euler–Maruyama discretization:

$$x_{t+h} = x_t - h\nabla V(x_t) + \sqrt{2h}\gamma, \qquad \gamma \sim \mathcal{N}(0, I)$$



Stochastic chain rule

$$\mathrm{d}f(g) = f'(g) \cdot \mathrm{d}g$$

- For an ODE $\mathrm{d}x_t = b_t(x_t)\mathrm{d}t$, for any differentiable f, we have

$$\mathrm{d}f(x_t) = \langle \nabla f(x_t), b_t(x_t) \rangle \mathrm{d}t$$

Itô's lemma. Let $f: \mathbb{R}^d \to \mathbb{R}$ be twice-differentiable, and suppose $\{\mathbf{x}_t\}_{t \geq 0}$ follows the SDE:

$$\mathrm{d}\mathbf{x}_t = b_t(\mathbf{x}_t)\mathrm{d}t + \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$

Then $\{f(\mathbf{x}_t)\}_{t \geq 0}$ is a stochastic process following the SDE:

$$\mathrm{d}f(\mathbf{x}_t) = \left(\langle \nabla f(\mathbf{x}_t), b_t(\mathbf{x}_t) \rangle + \frac{1}{2} \langle \nabla^2 f(\mathbf{x}_t), \sigma_t(\mathbf{x}_t)\sigma_t(\mathbf{x}_t)^\mathsf{T} \rangle \right) \mathrm{d}t + \langle \nabla f(\mathbf{x}_t), \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t \rangle$$

"$\mathrm{d}\boldsymbol{B}_t^2 = \mathrm{d}t$"

Stochastic chain rule

$$\mathrm{d}f(g) = f'(g) \cdot \mathrm{d}g$$

- For an ODE $\mathrm{d}x_t = b_t(x_t)\mathrm{d}t$, for any differentiable f, we have

$$\mathrm{d}f(x_t) = \langle \nabla f(x_t), b_t(x_t) \rangle \mathrm{d}t$$

Itô's lemma. Let $f: \mathbb{R}^d \to \mathbb{R}$ be twice-differentiable, and suppose $\{\mathbf{x}_t\}_{t \geq 0}$ follows the SDE:

$$\mathrm{d}\mathbf{x}_t = b_t(\mathbf{x}_t)\mathrm{d}t + \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$$

Then $\{f(\mathbf{x}_t)\}_{t \geq 0}$ is a stochastic process following the SDE:

$$\mathrm{d}f(\mathbf{x}_t) = \left(\langle \nabla f(\mathbf{x}_t), b_t(\mathbf{x}_t) \rangle + \frac{1}{2} \langle \nabla^2 f(\mathbf{x}_t), \sigma_t(\mathbf{x}_t)\sigma_t(\mathbf{x}_t)^\mathsf{T} \rangle \right) \mathrm{d}t + \langle \nabla f(\mathbf{x}_t), \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t \rangle$$

- Consider the Langevin diffusion: $\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\boldsymbol{B}_t$

$$\mathrm{d}f(\mathbf{x}_t) = (-\langle \nabla f, \nabla V \rangle + \Delta f)\mathrm{d}t + \sqrt{2}\langle \nabla f, \mathrm{d}\boldsymbol{B}_t \rangle, \qquad \Delta f \coloneqq \mathrm{tr}[\nabla^2 f]$$

Continuity equation

Given an ODE $\mathrm{d}x_t = b_t(x_t)\mathrm{d}t$ with $x_0 \sim q_0$, we get a flow of measures $q_t := \mathrm{law}(x_t)$

Continuity equation:

$$\partial_t q_t = -\mathrm{div}(q_t b_t)$$

- Consider the simplest SDE: $\mathrm{d}\mathbf{x}_t = \mathrm{d}\boldsymbol{B}_t,\ q_t = \mathcal{N}(x_0, tI)$ (the noising process)

$$q_t(x) = \frac{\exp(-\|x - x_0\|^2/(2t))}{(2\pi t)^{d/2}}, \qquad \partial_t q_t(x) = q_t(x)\left(\frac{\|x - x_0\|^2}{2t^2} - \frac{d}{2t}\right)$$

$$\Delta q_t(x) = \mathrm{div}\left(-q_t(x)\frac{x - x_0}{t}\right) = q_t(x)\left(\frac{\|x - x_0\|^2}{t^2} - \frac{d}{t}\right)$$

$$\partial_t q_t = \frac{1}{2}\Delta q_t$$ heat equation

Continuity equation

Recall the deterministic sampler:

$$\tilde{x}_{t-h} = \tilde{x}_t + \frac{1}{2} h \nabla \log q_t(\tilde{x}_t)$$

- The corresponding ODE is:

$$\mathrm{d}x_t^{\leftarrow} = \frac{1}{2} \nabla \log q_{T-t}(x_t^{\leftarrow}) \, \mathrm{d}t$$

- Let $q_t^{\leftarrow} := \mathrm{law}(x_t^{\leftarrow})$. By the continuity equation,

$$\partial_t q_t^{\leftarrow} = -\frac{1}{2} \mathrm{div}(q_t^{\leftarrow} \nabla \log q_{T-t}), \qquad q_0^{\leftarrow} = q_T$$

- $q_t^{\leftarrow} = q_{T-t}$ is a solution:

$$-\frac{1}{2} \mathrm{div}(q_{T-t} \nabla \log q_{T-t}) = -\frac{1}{2} \mathrm{div}\left(q_{T-t} \frac{\nabla q_{T-t}}{q_{T-t}} \right) = -\frac{1}{2} \Delta q_{T-t}$$

reverse of heat equation

Dual views of PDEs induced by diffusion

PDE on probability measures q_t

$$\partial_t q_t = \cdots$$

PDE on observables

- $P_{s\to t} f(\mathbf{x}) \coloneqq \mathbb{E}[f(\mathbf{x}_t) \mid \mathbf{x}_s = \mathbf{x}] \qquad \text{for} \quad f\colon \mathbb{R}^d \to \mathbb{R}$
- Semigroup structure: $P_{s\to t} P_{t\to w} = P_{s\to w}$
- Define $P_t \coloneqq P_{0\to t}$ ("smoothing operator")

$$\partial_t P_t f = \cdots$$

Kolmogorov backward equation

$$P_t f(\mathbf{x}) \coloneqq \mathbb{E}[f(\mathbf{x}_t) \mid \mathbf{x}_0 = \mathbf{x}]$$

- Note that

$$\partial_t P_t f = \lim_{h \to 0} \frac{P_{t+h} f - P_t f}{h} = P_t \underbrace{\lim_{h \to 0} \frac{P_{t \to t+h} - \mathrm{id}}{h} f}_{\mathcal{L}_t}$$

$$\partial_t P_t f = P_t \mathcal{L}_t f \qquad \text{(KBE)}$$

- For a general SDE $\mathrm{d}\mathbf{x}_t = b_t(\mathbf{x}_t)\mathrm{d}t + \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t$, Ito's formula gives

$$\mathrm{d}f(\mathbf{x}_t) = \left(\langle \nabla f(\mathbf{x}_t), b_t(\mathbf{x}_t) \rangle + \frac{1}{2} \langle \nabla^2 f(\mathbf{x}_t), \sigma_t(\mathbf{x}_t)\sigma_t(\mathbf{x}_t)^\mathsf{T} \rangle \right) \mathrm{d}t + \underbrace{\langle \nabla f(\mathbf{x}_t), \sigma_t(\mathbf{x}_t)\mathrm{d}\boldsymbol{B}_t \rangle}_{\text{mean zero}}$$

- Taking expectation $\mathbb{E}[\cdot \mid \mathbf{x}_0 = \mathbf{x}]$ on both sides gives

$$\mathcal{L}_t f = \langle \nabla f, b_t \rangle + \frac{1}{2} \langle \nabla^2 f, \sigma_t \sigma_t^\mathsf{T} \rangle$$

Kolmogorov forward equation

$$P_t f(\mathbf{x}) \coloneqq \mathbb{E}[f(\mathbf{x}_t) \mid \mathbf{x}_0 = \mathbf{x}]$$

$$\int f \mathrm{d}q_t = \int P_t f \mathrm{d}q_0 = \int f \mathrm{d}(P_t^* q_0)$$

where P_t^* is the adjoint of P_t

$$\mathbb{E}[f(\mathbf{x}_t)] = \int \mathbb{E}[f(\mathbf{x}_t)|\mathbf{x}_0 = \mathbf{x}] q_0(\mathbf{x}) \mathrm{d}\mathbf{x} = \int P_t f(\mathbf{x}) q_0(\mathbf{x}) \mathrm{d}\mathbf{x} = \int f(\mathbf{x}) P_t^* q_0(\mathbf{x}) \mathrm{d}\mathbf{x}$$

$$\partial_t \int f \mathrm{d}q_t = \int P_t \mathcal{L}_t f \mathrm{d}q_0 = \int f \mathrm{d}(P_t \mathcal{L}_t)^* q_0 = \int f \mathrm{d}(\mathcal{L}_t^* q_t)$$

$$\partial_t q_t = \mathcal{L}_t^* q_t \qquad \text{(KFE)}$$

Kolmogorov forward equation

$$\partial_t q_t = \mathcal{L}_t^* q_t \qquad \text{(KFE)}$$

- Recall

$$\mathcal{L}_t f = \langle \nabla f, b_t \rangle + \frac{1}{2} \langle \nabla^2 f, \sigma_t \sigma_t^\top \rangle$$

$$\int f \mathrm{d}(\mathcal{L}_t^* q_t) = \int \mathcal{L}_t f \mathrm{d} q_t = \int \left(\langle \nabla f, b_t \rangle + \frac{1}{2} \langle \nabla^2 f, \sigma_t \sigma_t^\top \rangle \right) \mathrm{d} q_t$$

$$= \int f \mathrm{d} \Bigg(\underbrace{-\mathrm{div}(b_t q_t) + \frac{1}{2} \sum_{i,j \in [d]} \frac{\partial^2}{\partial \mathbf{x}_i \partial \mathbf{x}_j} \left((\sigma_t \sigma_t^\top)_{ij} q_t \right)}_{\mathcal{L}_t^* q_t} \Bigg)$$

$$\partial_t q_t = -\mathrm{div}(b_t q_t) + \frac{1}{2} \Delta(\sigma_t^2 q_t) \qquad \text{(Fokker} - \text{Planck)}$$

Fokker-Planck equation

Recall the stochastic sampler:

$$\tilde{x}_{t-h} = \tilde{x}_t + h\nabla \log q_t(\tilde{x}_t) + \sqrt{h}\gamma, \qquad \gamma \sim \mathcal{N}(0, I)$$

- The corresponding SDE is:

$$\mathbf{x}_t^{\leftarrow} = \nabla \log q_{T-t}(\mathbf{x}_t^{\leftarrow})\,\mathrm{d}t + \mathrm{d}\boldsymbol{B}_t$$

- Let $q_t^{\leftarrow} \coloneqq \mathrm{law}(x_t^{\leftarrow})$. By Fokker-Planck equation,

$$\partial_t q_t^{\leftarrow} = -\mathrm{div}(q_t^{\leftarrow}\nabla \log q_{T-t}) + \frac{1}{2}\Delta q_t^{\leftarrow}, \qquad q_0^{\leftarrow} = q_T$$

- $q_t^{\leftarrow} = q_{T-t}$ is a solution:

$$-\mathrm{div}(q_{T-t}\nabla \log q_{T-t}) + \frac{1}{2}\Delta q_{T-t} = -\frac{1}{2}\Delta q_{T-t}$$

reverse of heat equation

Fokker-Planck equation

Recall the Langevin diffusion:

$$\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\boldsymbol{B}_t$$

- By Fokker-Planck equation,

$$\partial_t q_t = \mathrm{div}(q_t \nabla V) + \Delta q_t$$

- $q = \frac{1}{Z} e^{-V}$ is a solution:

$$\mathrm{div}(q\nabla V) + \Delta q = -\mathrm{div}(\nabla q) + \Delta q = -\Delta q + \Delta q = 0$$

Spectral gap and the Dirichlet form

- **Motivation:** we want to understand the spectrum of $\mathcal{L}^*$

$$\frac{\partial}{\partial t}\pi_t = \mathcal{L}^*\pi_t \quad \Longrightarrow \quad \pi_t = \exp(t\mathcal{L}^*)\pi_0$$

If $\pi^\star$ is an eigenfunction of $\mathcal{L}^*$ with eigenvalue 0 and all other eigenvalues are negative

Then $\pi_t \longrightarrow \pi^\star$ as $t \longrightarrow \infty$

- To quantitively analyze the convergence rate, we define the Dirichlet form:

$$\mathcal{E}(f,g) \coloneqq -\int f(\mathbf{x})\mathcal{L}g(\mathbf{x})\pi^\star(\mathbf{x})\mathrm{d}\mathbf{x} \qquad \forall\, f,g\colon \mathbb{R}^d \to \mathbb{R}$$

- For the Langevin diffusion, $-\langle f, \mathcal{L}g\rangle_{\pi^\star}$

$$\mathcal{E}(f,g) \coloneqq \int \langle \nabla f(\mathbf{x}), \nabla g(\mathbf{x})\rangle \pi^\star(\mathbf{x})\mathrm{d}\mathbf{x}$$

Courant-Fischer:

$$\frac{v^\mathsf{T}\boldsymbol{A}v}{v^\mathsf{T}v}$$

Dirichlet form

$$\mathcal{E}(f,g) \coloneqq \int \langle \nabla f(\mathbf{x}), \nabla g(\mathbf{x}) \rangle \pi^\star(\mathbf{x}) \mathrm{d}\mathbf{x}$$

- Langevin is reversible since $\mathcal{E}(f,g) = \mathcal{E}(g,f)$ $\qquad \langle \mathcal{L}f, g \rangle_{\pi^\star} = \langle f, \mathcal{L}g \rangle_{\pi^\star}$
- Constant function is an eigenfunction of $\mathcal{E}$ with eigenvalue 0
- The spectral gap is captured by all eigenfunctions of $\mathcal{E}$ orthogonal to the constant function

$$\mathrm{gap} \coloneqq \min_{f \perp 1} \frac{\mathcal{E}(f,f)}{\langle f, f \rangle_{\pi^\star}}$$

$$\mathcal{E}(f,g) = -\int \mathcal{L}f(\mathbf{x}) g(\mathbf{x}) \pi^\star(\mathbf{x}) \mathrm{d}\mathbf{x} = \int \langle \nabla f, \nabla V \rangle g \pi^\star \mathrm{d}\mathbf{x} - \int \Delta f g \pi^\star \mathrm{d}\mathbf{x}$$

(I.B.P.)

$$= \int \langle \nabla f, \nabla V \rangle g \pi^\star \mathrm{d}\mathbf{x} + \left(\int \langle \nabla f, \nabla g \rangle \pi^\star \mathrm{d}\mathbf{x} + \int \langle \nabla f, \nabla \pi^\star \rangle g \mathrm{d}\mathbf{x} \right) \qquad \pi^\star \propto \exp(-V)$$

$$= \int \langle \nabla f, \nabla V \rangle g \pi^\star \mathrm{d}\mathbf{x} + \int \langle \nabla f, \nabla g \rangle \pi^\star \mathrm{d}\mathbf{x} - \int \langle \nabla f, \nabla V \rangle g \pi^\star \mathrm{d}\mathbf{x} = \int \langle \nabla f, \nabla g \rangle \pi^\star \mathrm{d}\mathbf{x}$$

Today's plan

1. Drift-diffusion processes
2. Markov semigroup
3. Optimal transport
4. Functional inequalities

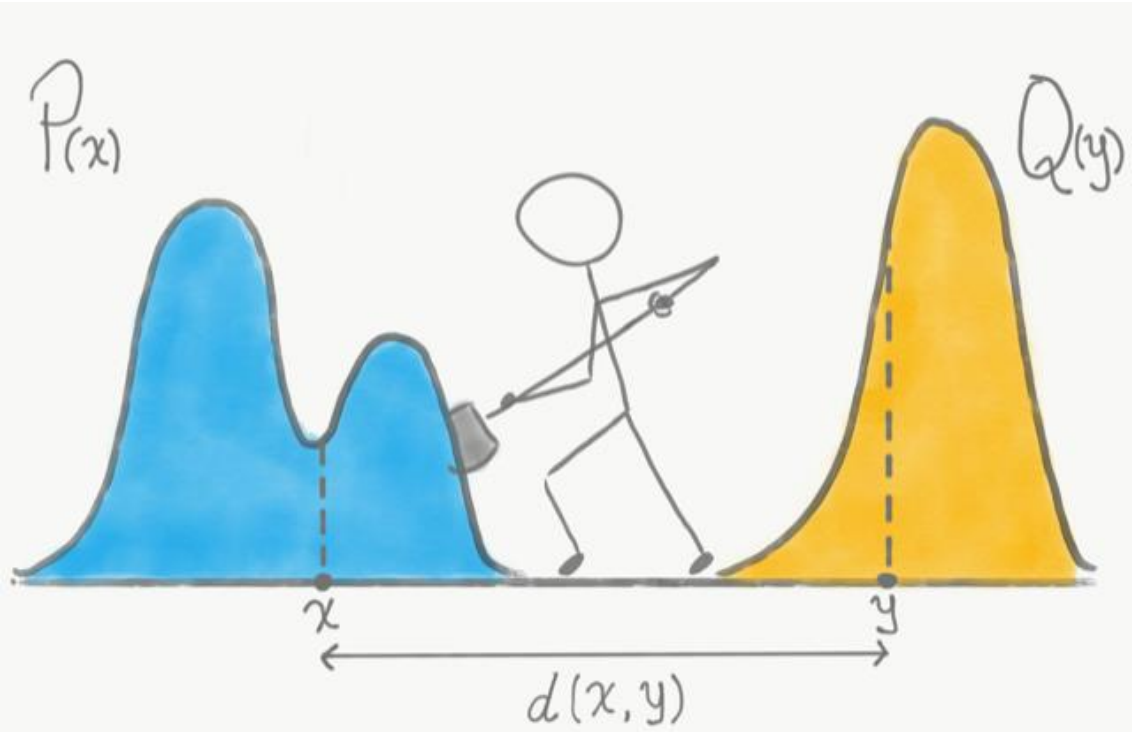
Highly recommend reference: *Log-concave sampling* by Sinho Chewi '25

Optimal transport

Wasserstein distance:

$$W_2(\mu,\pi) := \inf_{\gamma \in \mathcal{C}(\mu,\pi)} \sqrt{\int \|x-y\|^2 \gamma(x,y)\mathrm{d}x\mathrm{d}y}$$

γ is a coupling of μ and ν



Optimal transport

Wasserstein distance:

$$W_2(\mu,\pi) \coloneqq \inf_{\gamma\in\mathcal{C}(\mu,\pi)} \sqrt{\int \|x-y\|^2 \gamma(x,y)\mathrm{d}x\mathrm{d}y}$$

Kantorovich: By the strong duality of LP,

$$W_2(\mu,\pi) \coloneqq \sup_{(f,g)\in\mathcal{D}(\mu,\pi)} \int f(x)\mu(x)\mathrm{d}x + \int g(y)\nu(y)\mathrm{d}y$$

$$\mathcal{D}(\mu,\nu) \coloneqq \left\{(f,g)\in L^1 \;\middle|\; f(x)+g(y) \le \frac{1}{2}\|x-y\|^2 \;\; \forall\, x,y\in\mathbb{R}^d\right\}$$

Brenier: the optimal coupling is $x\sim\mu$ and $y\leftarrow\nabla\phi(x)$ for some convex potential ϕ

Wasserstein convergence of Langevin dynamics

- $\pi \propto \exp(-V)$ is μ-strongly logconcave if V is μ-strongly convex (i.e., $\nabla^2 V \succcurlyeq \mu I$)

Theorem. Let $\{\mathbf{x}_t\}_{t\geq 0}$ follow the Langevin diffusion with stationary distribution $\pi^\star$ being μ-strongly logconcave. Then for all $t \geq 0$,

$$W_2^2(\pi_t, \pi) \leq \exp(-2\mu t)\, W_2^2(\pi_0, \pi^\star)$$

Proof.

- We use the "coupling method":
  - $\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\boldsymbol{B}_t$
  - $\mathrm{d}\mathbf{x}_t^\star = -\nabla V(\mathbf{x}_t^\star)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\boldsymbol{B}_t$

  same

  $(\mathbf{x}_0, \mathbf{x}_0^\star) \sim \gamma_0$ the optimal coupling for $W_2(\pi_0, \pi^\star)$
- We get a coupling γ_t for $(\pi_t, \pi^\star)$ for every $t \geq 0$

Wasserstein convergence of Langevin dynamics

- $\pi \propto \exp(-V)$ is μ-strongly logconcave if V is μ-strongly convex (i.e., $\nabla^2 V \succcurlyeq \mu I$)

Theorem. Let $\{\mathbf{x}_t\}_{t\geq 0}$ follow the Langevin diffusion with stationary distribution $\pi^\star$ being μ-strongly logconcave. Then for all $t \geq 0$,

$$W_2^2(\pi_t, \pi) \leq \exp(-2\mu t)\, W_2^2(\pi_0, \pi^\star)$$

Proof.

- We use the "coupling method":
  - $\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\boldsymbol{B}_t$
  - $\mathrm{d}\mathbf{x}_t^\star = -\nabla V(\mathbf{x}_t^\star)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\boldsymbol{B}_t$ (same $\boldsymbol{B}_t$)
- We get a coupling γ_t for $(\pi_t, \pi^\star)$ for every $t \geq 0$

$$\frac{\mathrm{d}}{\mathrm{d}s}(\mathbf{x}_s - \mathbf{x}_s^\star) = \nabla V(\mathbf{x}_s^\star) - \nabla V(\mathbf{x}_s)$$

$$\Longrightarrow \frac{\mathrm{d}}{\mathrm{d}s}\|\mathbf{x}_s - \mathbf{x}_s^\star\|^2 = -2\langle \nabla V(\mathbf{x}_s) - \nabla V(\mathbf{x}_s^\star), \mathbf{x}_s - \mathbf{x}_s^\star\rangle$$

$$\leq -2\mu\|\mathbf{x}_s - \mathbf{x}_s^\star\|^2$$

by the strong convexity of V

Wasserstein convergence of Langevin dynamics

- $\pi \propto \exp(-V)$ is μ-strongly logconcave if V is μ-strongly convex (i.e., $\nabla^2 V \succcurlyeq \mu I$)

Theorem. Let $\{\mathbf{x}_t\}_{t\geq 0}$ follow the Langevin diffusion with stationary distribution $\pi^\star$ being μ-strongly logconcave. Then for all $t \geq 0$,

$$W_2^2(\pi_t, \pi) \leq \exp(-2\mu t)\, W_2^2(\pi_0, \pi^\star)$$

Proof.

- We use the "coupling method":
  - $\mathrm{d}\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\boldsymbol{B}_t$
  - $\mathrm{d}\mathbf{x}_t^\star = -\nabla V(\mathbf{x}_t^\star)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\boldsymbol{B}_t$ (same $\boldsymbol{B}_t$)
- We get a coupling γ_t for $(\pi_t, \pi^\star)$ for every $t \geq 0$

$$\frac{\mathrm{d}}{\mathrm{d}s}(\mathbf{x}_s - \mathbf{x}_s^\star) = \nabla V(\mathbf{x}_s^\star) - \nabla V(\mathbf{x}_s)$$

$$\Longrightarrow \frac{\mathrm{d}}{\mathrm{d}s}\|\mathbf{x}_s - \mathbf{x}_s^\star\|^2 = -2\langle \nabla V(\mathbf{x}_s) - \nabla V(\mathbf{x}_s^\star), \mathbf{x}_s - \mathbf{x}_s^\star\rangle$$

$$\leq -2\mu\|\mathbf{x}_s - \mathbf{x}_s^\star\|^2$$

$$\Longrightarrow \|\mathbf{x}_t - \mathbf{x}_t^\star\|^2 \leq \exp(-2\mu t)\,\|\mathbf{x}_0 - \mathbf{x}_0^\star\|^2$$

by Gronwall's inequality

Wasserstein convergence of Langevin dynamics

- $\pi \propto \exp(-V)$ is μ-strongly logconcave if V is μ-strongly convex (i.e., $\nabla^2 V \succcurlyeq \mu I$)

Theorem. Let $\{\mathbf{x}_t\}_{t\geq 0}$ follow the Langevin diffusion with stationary distribution $\pi^\star$ being μ-strongly logconcave. Then for all $t \geq 0$,

$$W_2^2(\pi_t, \pi) \leq \exp(-2\mu t)\, W_2^2(\pi_0, \pi^\star)$$

Proof.

- Since W_2 minimizes over all couplings, we have

$$\begin{aligned} W_2^2(\pi_t, \pi^\star) \leq \mathbb{E}_{(\mathbf{x}_t, \mathbf{x}_t^\star)\sim\gamma_t}[\|\mathbf{x}_t - \mathbf{x}_t^\star\|^2] &\leq \exp(-2\mu t)\, \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_0^\star)\sim\gamma_0}[\|\mathbf{x}_0 - \mathbf{x}_0^\star\|^2] \\ &= \exp(-2\mu t)\, W_2(\pi_0, \pi^\star)^2 \end{aligned}$$

■

Today's plan

1. Drift-diffusion processes
2. Markov semigroup
3. Optimal transport
4. Functional inequalities

Highly recommend reference: *Log-concave sampling* by Sinho Chewi '25

Poincaré inequality

We say that $\pi^\star$ satisfies a Poincaré inequality with constant C_{PI} if for all differentiable f,

$$\mathrm{Var}_{\pi^\star}[f] \leq C_{\mathrm{PI}} \int \|\nabla f(\mathbf{x})\|^2 \pi^\star(\mathbf{x}) \mathrm{d}\mathbf{x}$$

- Poincaré inequality is equivalent to the statement that $\langle f, f \rangle_{\pi^\star}$ $\mathcal{E}(f, f)$

$$\mathrm{gap} := \min_{f \perp 1} \frac{\mathcal{E}(f,f)}{\langle f, f \rangle_{\pi^\star}} \geq \frac{1}{C_{\mathrm{PI}}}$$

- The χ^2 divergence between μ and ν is defined as:

$$\chi^2(\mu \| \nu) := \int \left(\frac{\mu(x)}{\nu(x)} \right)^2 \nu(x) \mathrm{d}x - 1$$

- PI $\Longrightarrow$ convergence in χ^2 divergence

Convergence from Poincaré inequality

Theorem. Let $\{\mathbf{x}_t\}_{t\geq 0}$ follow the Langevin diffusion with stationary distribution $\pi^\star$ satisfying a Poincaré inequality with constant C_{PI}. Then for all $t \geq 0$,

$$\chi^2(\pi_t \| \pi^\star) \leq \exp\left(-\frac{2t}{C_{\mathrm{PI}}}\right) \chi^2(\pi_0 \| \pi^\star)$$

Proof.

$$\frac{\mathrm{d}}{\mathrm{d}t}\chi^2(\pi_t \| \pi^\star) = \frac{\mathrm{d}}{\mathrm{d}t}\int \left(\frac{\pi_t(\mathbf{x})^2}{\pi^\star(\mathbf{x})^2} - 1\right)\pi^\star(\mathbf{x})\mathrm{d}\mathbf{x} = 2\int \left(\frac{\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right)\left(\frac{\partial}{\partial t}\frac{\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right)\pi^\star(\mathbf{x})\mathrm{d}\mathbf{x}$$

$$= 2\int \left(\frac{\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right)\left(\frac{\mathcal{L}^*\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right)\pi^\star(\mathbf{x})\mathrm{d}\mathbf{x} = -2\mathcal{E}\left(\frac{\pi_t}{\pi^\star}, \frac{\pi_t}{\pi^\star}\right)$$

$$\leq -\frac{2}{C_{\mathrm{PI}}}\mathrm{Var}_{\pi^\star}\left[\frac{\pi_t}{\pi^\star}\right] = -\frac{2}{C_{\mathrm{PI}}}\chi^2(\pi_t \| \pi^\star)$$

■

Log-Sobolev inequality

We say that $\pi^\star$ satisfies a log-Sobolev inequality with constant C_{LSI} if for all differentiable f,

$$\mathrm{Ent}_{\pi^\star}[f^2] \leq 2C_{\mathrm{LSI}} \int \|\nabla f(\mathbf{x})\|^2 \pi^\star(\mathbf{x}) \mathrm{d}\mathbf{x}$$

where $\mathrm{Ent}_{\pi^\star}[f] \coloneqq \mathbb{E}_{\pi^\star}[f \log f] - \mathbb{E}_{\pi^\star}[f] \log \mathbb{E}_{\pi^\star}[f]$.

Equivalently,

$$D_{\mathrm{KL}}\left(\pi \middle\| \pi^\star\right) \leq \frac{C_{\mathrm{LSI}}}{2} \int \left\| \nabla \log \frac{\pi(\mathbf{x})}{\pi^\star(\mathbf{x})} \right\|^2 \pi(\mathbf{x}) \mathrm{d}\mathbf{x}$$

where $D_{\mathrm{KL}}\left(\pi \middle\| \pi^\star\right) \coloneqq \int \pi(\mathbf{x}) \log\left(\frac{\pi(\mathbf{x})}{\pi^\star(\mathbf{x})}\right) \mathrm{d}\mathbf{x}$

Fisher information

- LSI $\Longrightarrow$ convergence in KL divergence

Convergence from log-Sobolev inequality

Theorem. Let $\{\mathbf{x}_t\}_{t\geq 0}$ follow the Langevin diffusion with stationary distribution $\pi^\star$ satisfying a log-Sobolev inequality with constant C_{LSI}. Then for all $t \geq 0$,

$$D_{\mathrm{KL}}(\pi_t \| \pi^\star) \leq \exp\left(-\frac{2t}{C_{\mathrm{LSI}}}\right) D_{\mathrm{KL}}(\pi_0 \| \pi^\star)$$

Proof.

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\pi_t \| \pi^\star) = \frac{\mathrm{d}}{\mathrm{d}t} \int \pi_t(\mathbf{x}) \log\left(\frac{\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right) \mathrm{d}\mathbf{x} = \int \log\left(\frac{\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right) \left(\frac{\mathcal{L}^* \pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right) \pi^\star(\mathbf{x}) \mathrm{d}\mathbf{x}$$

$$= -\mathcal{E}\left(\frac{\pi_t}{\pi^\star}, \log\left(\frac{\pi_t}{\pi^\star}\right)\right) = -\int \left\langle \nabla \log\left(\frac{\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right), \nabla\left(\frac{\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right) \right\rangle \pi^\star(\mathbf{x}) \mathrm{d}\mathbf{x}$$

$$= -\int \left\| \nabla \log\left(\frac{\pi_t(\mathbf{x})}{\pi^\star(\mathbf{x})}\right) \right\|^2 \pi^\star(\mathbf{x}) \mathrm{d}\mathbf{x} \leq -\frac{2}{C_{\mathrm{LSI}}} D_{\mathrm{KL}}(\pi_t \| \pi^\star)$$

■

LSI vs PI convergence

$$\chi^2(\pi_t \| \pi^\star) \leq \exp\left(-\frac{2t}{C_{\mathrm{PI}}}\right) \chi^2(\pi_0 \| \pi^\star)$$

$$D_{\mathrm{KL}}(\pi_t \| \pi^\star) \leq \exp\left(-\frac{2t}{C_{\mathrm{LSI}}}\right) D_{\mathrm{KL}}(\pi_0 \| \pi^\star)$$

LSI provides stronger convergence guarantee than PI since KL divergence can be exponentially smaller than χ^2 divergence

- π_0 is a β-warm start w.r.t. $\pi^\star$ if $\frac{\pi_0(\mathbf{x})}{\pi^\star(\mathbf{x})} \leq \beta$ for all $\mathbf{x} \in \mathbb{R}^d$
- $\chi^2(\pi_0 \| \pi^\star) \leq \beta^2$
- $D_{\mathrm{KL}}(\pi_0 \| \pi^\star) = \mathbb{E}_{\pi_0}[\log(\pi_0/\pi^\star)] \leq \log \beta$

LSI and PI constants

Lemma. If π satisfies a log-Sobolev inequality with constant C, it also satisfies a Poincaré inequality with constant C.

How to determine the LSI or PI constant?

$$\mu\text{-strongly logconcave} \implies C_{\mathrm{PI}} = \frac{1}{\mu} \quad \text{(Brascamp-Lieb inequlity)}$$

$$\implies C_{\mathrm{LSI}} = \frac{1}{\mu} \quad \text{(Bakry–Émery thoerem)}$$

LSI / PI implies concentration

Lemma. Let $\pi^\star$ satisfies a log-Sobolev inequality with constant C_{LSI}. Then for any 1-Lipschitz function $f\colon \mathbb{R}^d \to \mathbb{R}$, we have that

$$\Pr_{\mathbf{x}\sim\pi^\star}[f(\mathbf{x}) \geq \mathbb{E}_{\pi^\star}[f] + \epsilon] \leq \exp\left(-\frac{\epsilon^2}{2C_{\mathrm{LSI}}}\right) \quad \forall\, \epsilon > 0$$

sub-Gaussian

Lemma. Let $\pi^\star$ satisfies a Poincaré inequality with constant C_{PI}. Then for any 1-Lipschitz function $f\colon \mathbb{R}^d \to \mathbb{R}$, we have that

$$\Pr_{\mathbf{x}\sim\pi^\star}[f(\mathbf{x}) \geq \mathbb{E}_{\pi^\star}[f] + \epsilon] \leq 3\exp\left(-\frac{\epsilon}{\sqrt{C_{\mathrm{PI}}}}\right) \quad \forall\, \epsilon > 0$$

sub-exponential